

Hypertension-Cardiology Data Challenge

Predicting BP Control Failure and Evaluating Drug Outcomes Across Comorbidity

Project Background

This project was completed as part of the Zion Tech Hub Data Challenge, analyzing a Hypertension Cardiology Center dataset of 350 patient records. The goal was to analyze patient outcomes, identify risk factors for poor blood pressure control, and recommend better treatment strategies.

Business Problem

The challenge centered on two core questions: what predicts BP control failure at 3 months, and can high-risk patients be identified at the point of prescription? And which anti-hypertensive drug class delivers the best outcomes for patients with specific comorbidity profiles?

Project Objectives

- Identify predictors of BP control failure at the 3-month mark.
- Determine the best-performing drug class for each comorbidity profile.
- Evaluate demographic relationships (age, gender) with treatment outcomes.
- Build a predictive model to flag high-risk patients at prescription.
- Translate findings into actionable treatment recommendations.

Key Insights

- 1. Overall & Demographic Patterns:** Of 350 patients, 317 did not achieve BP control after 3 months. Age alone was not the dominant driver as control rates ranged from 100% in the 80–89 group down to 85.25% in the 40–49 group, without a clean linear pattern. Gender showed a clearer signal: Females had a higher control rate (11.24%) than Males (7.56%), linked to stronger medication adherence among female patients. Patients with existing comorbidities consistently showed higher BP failure rates than those without.
- 2. Drug Class Performance:** Across the full patient population, Combination Therapy led with a 21.05% control rate, followed by Thiazide Diuretic (13.11%), Beta Blocker (9.84%), ARB (5.88%), and Calcium Channel Blocker (5.17%) with ACE Inhibitor at 0%. Effectiveness shifted by comorbidity: Thiazide Diuretic performed best for chronic kidney disease (22.22%), while Combination Therapy led for Diabetes (25%), Dyslipidemia (40%), and Obesity (25%). Across every comorbidity profile whether single or combined, Combination Therapy delivered the most consistent results, suggesting patients with complex health conditions benefit more from combined therapies than single-drug treatment.
- 3. Predictive Modeling & Risk Identification:** A predictive model was built using features available at the point of prescription; age, gender, comorbidities, baseline BP, drug class,

adherence, and visit count. The model achieved 92.86% accuracy, a recall of 1.00 for BP failure, and a ROC-AUC of 0.95, meaning it correctly flagged every patient in the test set who went on failing BP control. Patients were grouped into High, Moderate, and Low risk tiers based on predicted failure probability. The four strongest predictors of BP control failure were baseline BP severity, medication adherence, comorbidities, and antihypertensive drug class.

Recommendations

- **Treatment Strategy:** Treat severe baseline hypertension as high-risk from the point of prescription. Use personalized treatment plans matched to each patient's comorbidity profile rather than a one-size-fits-all approach and consider Combination Therapy as the default for patients with multiple risk factors.
- **Adherence & Follow-Up:** Strengthen medication adherence programs through counseling, reminders, and patient education, given how strongly adherence influences outcomes. Schedule closer, more frequent follow-up visits for patients identified as high-risk.
- **Predictive Risk Scoring:** Deploy the predictive model to flag high-risk patients early, allowing clinicians to intervene before treatment failure occurs, and prioritize monitoring resources toward the patients the model identifies as highest risk.

Limitation

The dataset's small sample size (350 patients) limits generalization. The predictive model should be validated against larger datasets before clinical deployment.

Conclusion

This project demonstrates that data analytics can help hypertension centers move from reactive care to risk-based patient management. Baseline BP severity, medication adherence, comorbidities, and drug class emerged as the key drivers of 3-month BP control outcomes, and Combination Therapy showed the most consistent performance across comorbidity profiles — pointing toward more personalized, proactive treatment strategies for complex patients.